Supplementary Theory and Algorithm Details for exdqlm

Working draft

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Abstract

This document records the model hierarchies, augmented posterior targets, MCMC updates, variational approximations, and ELBO components underlying the main dynamic models and the static ridge/RHS-NS paths in the current exdqlm implementation. It is written as a working supplement draft: sections that have already been checked against the existing theory notes are presented in mathematical form, and the final verification tasks are listed at the end. Vectors are written in bold lowercase, matrices in bold uppercase, and dimensions are given when each object is introduced.

Contents

1 Notation and Distributional Building Blocks

Let $p_0 \in (0,1)$ be the target quantile. The response is scalar. In dynamic models, observations are indexed by $t = 1, \ldots, T$; in static regression, observations are indexed by $i = 1, \ldots, n$. Lowercase bold symbols denote vectors, for example $\boldsymbol{\theta}_t \in \mathbb{R}^q$ and $\boldsymbol{\beta} \in \mathbb{R}^p$. Uppercase bold symbols denote matrices, for example $\mathbf{G}_t \in \mathbb{R}^{q\times q}$ and $\mathbf{X} \in \mathbb{R}^{n\times p}$. For any positive definite matrix $\mathbf{A}$, $|\mathbf{A}|$ denotes its determinant.

1.1 Notation Contract

The supplement uses the following conventions throughout. The symbols p_γ, A_γ, B_γ, and C_γ are deterministic functions of (p_0, γ), whereas $\pi_\gamma(\gamma)$ denotes the prior density for γ on (L, U). Generic densities are denoted by $p(\cdot)$ only when no confusion with the exAL quantile map can arise. In variational sections, expectations without an explicit subscript are taken under the current mean-field distribution q. The symbol ξ is reserved for the RHS-NS half-Cauchy auxiliary variable; the transformed coordinate for γ in Laplace-Delta calculations is denoted by z_γ.

Symbol	Meaning
p_0	Target quantile level.
p_γ	exAL internal quantile map determined by (p_0, γ).
A_0, B_0	AL constants $A(p_0)$ and $B(p_0)$.
$A_\gamma, B_\gamma, C_\gamma$	exAL constants evaluated at p_γ.
$\pi_\gamma(\gamma)$	Prior density for γ on (L, U).
$\boldsymbol{\theta}_t, \boldsymbol{f}_t, \mathbf{G}_t, \mathbf{W}_t$	Dynamic state, observation vector, evolution matrix, and evolution covariance.
$\boldsymbol{\beta}, \mathbf{X}$	Static regression coefficients and design matrix.
$\mathcal{A}$	Active coefficient set receiving RHS-NS shrinkage.

The distributional parameterizations are also fixed throughout: $\mathcal{N}(m, V)$ uses variance V; $\mathrm{IG}(a, b)$ uses rate b with density proportional to $x^{-(a+1)}\exp(-b/x)$; $\mathrm{Exp}(\mathrm{rate} = 1/\sigma)$ has density $\sigma^{-1}\exp(-x/\sigma)$; and $\mathcal{N}^+(m, V)$ denotes $\mathcal{N}(m, V)$ truncated to $(0, \infty)$. The generalized inverse Gaussian parameterization is given in (8).

The asymmetric Laplace (AL) mixture representation used by the package is

$$v \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma), \qquad v > 0, \tag{1}$$

$$y \mid \eta, \sigma, v \sim \mathcal{N}\{\eta + A(p_0)v,\ \sigma B(p_0)v\}, \tag{2}$$

where

$$A(p) = \frac{1-2p}{p(1-p)}, \qquad B(p) = \frac{2}{p(1-p)}. \tag{3}$$

The AL model is the $\gamma = 0$ special case of the exAL formulation below.

The exAL distribution is represented through an additional latent variable $s \sim \mathcal{N}^+(0, 1)$, a standard normal truncated to $(0, \infty)$. For $\gamma \in (L, U)$, define

$$g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2), \tag{4}$$

$$p_\gamma = \mathbb{I}(\gamma < 0) + \frac{p_0 - \mathbb{I}(\gamma < 0)}{g(\gamma)}, \tag{5}$$

$$A_\gamma = A(p_\gamma), \qquad B_\gamma = B(p_\gamma), \qquad C_\gamma = \{\mathbb{I}(\gamma > 0) - p_\gamma\}^{-1}. \tag{6}$$

The endpoints $L < 0 < U$ are the roots $g(L) = 1 - p_0$ and $g(U) = p_0$. Conditional on (v, s, σ, γ),

$$y \mid \eta, \sigma, \gamma, v, s \sim \mathcal{N}\left(\eta + C_\gamma \sigma |\gamma| s + A_\gamma v,\ \sigma B_\gamma v\right). \tag{7}$$

The package uses an inverse-gamma prior $\sigma \sim \mathrm{IG}(a_\sigma, b_\sigma)$ and a proper truncated Student-t prior $\pi_\gamma(\gamma)$ for γ on (L, U).

The generalized inverse Gaussian distribution is parameterized as

$$x \sim \mathrm{GIG}(\lambda, \chi, \psi) \quad \iff \quad p(x) \propto x^{\lambda-1} \exp\left\{-\frac{1}{2}\left(\frac{\chi}{x} + \psi x\right)\right\}, \qquad x > 0. \tag{8}$$

For $r \in \mathbb{R}$,

$$\mathbb{E}(x^r) = \left(\frac{\chi}{\psi}\right)^{r/2} \frac{K_{\lambda+r}(\sqrt{\chi\psi})}{K_\lambda(\sqrt{\chi\psi})}, \tag{9}$$

where $K_\nu(\cdot)$ is the modified Bessel function of the second kind.

2 Dynamic DQLM under the AL Likelihood

2.1 Model Hierarchy and Full Joint

Let $\boldsymbol{\theta}_t \in \mathbb{R}^q$ be the latent state, $\boldsymbol{f}_t \in \mathbb{R}^q$ the observation vector, $\mathbf{G}_t \in \mathbb{R}^{q\times q}$ the evolution matrix, and $\mathbf{W}_t \in \mathbb{R}^{q\times q}$ a positive semidefinite evolution covariance. The dynamic DQLM is

$$\boldsymbol{\theta}_0 \sim \mathcal{N}(\boldsymbol{m}_0, \mathbf{C}_0), \qquad \boldsymbol{m}_0 \in \mathbb{R}^q,\ \mathbf{C}_0 \in \mathbb{R}^{q\times q}, \tag{10}$$

$$\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t), \tag{11}$$

$$v_t \mid \sigma \sim \mathrm{Exp}(\text{rate} = 1/\sigma), \tag{12}$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, v_t \sim \mathcal{N}(\boldsymbol{f}_t^\top \boldsymbol{\theta}_t + A_0 v_t,\ \sigma B_0 v_t), \tag{13}$$

where $A_0 = A(p_0)$ and $B_0 = B(p_0)$. With $\boldsymbol{v} = (v_1, \ldots, v_T)^\top$ and $\boldsymbol{\theta}_{0:T} = (\boldsymbol{\theta}_0, \ldots, \boldsymbol{\theta}_T)$, the posterior target is

$$p(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v} \mid \boldsymbol{y}) \propto p(\sigma)\, p(\boldsymbol{\theta}_0) \prod_{t=1}^{T} p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^{T} p(v_t \mid \sigma) \prod_{t=1}^{T} p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t). \tag{14}$$

Equivalently, conditional on $(\sigma, \boldsymbol{v})$, the pseudo-observation $\tilde{y}_t = y_t - A_0 v_t$ satisfies

$$\tilde{y}_t = \boldsymbol{f}_t^\top \boldsymbol{\theta}_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma B_0 v_t). \tag{15}$$

Thus the conditional state posterior is a Gaussian state-space posterior.

2.2 MCMC Updates

One Gibbs sweep for the dynamic DQLM uses the following full conditionals.

State path. Given $(\sigma, \boldsymbol{v})$, sample $\boldsymbol{\theta}_{0:T}$ exactly by forward-filtering backward-sampling (FFBS) applied to (15).

Scale. Let $r_t = y_t - \boldsymbol{f}_t^\top \boldsymbol{\theta}_t$. Then

$$\sigma \mid \boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{y} \sim \mathrm{IG}\left(a_\sigma + \frac{3T}{2},\ b_\sigma + \sum_{t=1}^{T} v_t + \sum_{t=1}^{T} \frac{(r_t - A_0 v_t)^2}{2B_0 v_t}\right). \tag{16}$$

Latent AL variables. For $t = 1, \dots, T$,

$$v_t \mid \boldsymbol{\theta}_t, \sigma, y_t \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{r_t^2}{\sigma B_0},\ \frac{A_0^2}{\sigma B_0} + \frac{2}{\sigma}\right). \tag{17}$$

2.3 Mean-field VB and ELBO

The dynamic DQLM VB approximation uses

$$q(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v}) = q(\boldsymbol{\theta}_{0:T}) q(\sigma) \prod_{t=1}^{T} q(v_t), \tag{18}$$

with $q(\sigma) = \mathrm{IG}(a_q, b_q)$ and $q(v_t) = \mathrm{GIG}(1/2, \chi_t, \psi_t)$. Define

$$\kappa = \mathbb{E}_q(\sigma^{-1}) = \frac{a_q}{b_q}, \qquad \ell_t = \mathbb{E}_q(v_t^{-1}), \qquad \nu_t = \mathbb{E}_q(v_t). \tag{19}$$

The state factor is the Gaussian posterior of the auxiliary model

$$y_t^\star = \boldsymbol{f}_t^\top \boldsymbol{\theta}_t + \epsilon_t^\star, \qquad \epsilon_t^\star \sim \mathcal{N}(0, R_t^\star), \qquad y_t^\star = y_t - \frac{A_0}{\ell_t}, \quad R_t^\star = \frac{B_0}{\kappa \ell_t}. \tag{20}$$

After Kalman smoothing, write $q(\boldsymbol{\theta}_t) = \mathcal{N}(\boldsymbol{m}_{t|T}, \mathbf{C}_{t|T})$ and define

$$\bar{r}_t = y_t - \boldsymbol{f}_t^\top \boldsymbol{m}_{t|T}, \tag{21}$$

$$\overline{r_t^2} = \bar{r}_t^2 + \boldsymbol{f}_t^\top \mathbf{C}_{t|T} \boldsymbol{f}_t. \tag{22}$$

The remaining CAVI updates are

$$q(v_t) = \mathrm{GIG}\left(\frac{1}{2},\ \frac{\kappa}{B_0}\overline{r_t^2},\ \kappa\left(2 + \frac{A_0^2}{B_0}\right)\right), \tag{23}$$

$$q(\sigma) = \mathrm{IG}(a_q, b_q), \qquad a_q = a_\sigma + \frac{3T}{2}, \tag{24}$$

$$b_q = b_\sigma + \sum_{t=1}^{T} \nu_t + \frac{1}{2B_0} \sum_{t=1}^{T} \left\{ \ell_t \overline{r_t^2} - 2A_0 \bar{r}_t + A_0^2 \nu_t \right\}. \tag{25}$$

The ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\boldsymbol{y}, \boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v})\} - \mathbb{E}_q\{\log q(\boldsymbol{\theta}_{0:T}, \sigma, \boldsymbol{v})\} \tag{26}$$

$$= \mathcal{L}_\theta + \mathbb{E}_q\{\log p(\sigma)\} + \sum_{t=1}^{T} \mathbb{E}_q\{\log p(v_t \mid \sigma)\} + \sum_{t=1}^{T} \mathbb{E}_q\{\log p(y_t \mid \boldsymbol{\theta}_t, \sigma, v_t)\} \tag{27}$$

$$+ H\{q(\sigma)\} + \sum_{t=1}^{T} H\{q(v_t)\}. \tag{28}$$

The state-path term $\mathcal{L}_\theta$ is evaluated from the auxiliary Gaussian state-space model in (20) through the Kalman marginal likelihood identity used in the package ELBO diagnostics.

3 Dynamic exDQLM under the exAL Likelihood

3.1 Model Hierarchy and Full Joint

The dynamic exDQLM replaces the AL observation in (13) with the exAL augmentation

$$v_t \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma), \qquad s_t \sim \mathcal{N}^+(0, 1), \tag{29}$$

$$y_t \mid \boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \mathcal{N}\left(\boldsymbol{f}_t^\top \boldsymbol{\theta}_t + C_\gamma \sigma |\gamma| s_t + A_\gamma v_t,\ \sigma B_\gamma v_t\right), \tag{30}$$

with the same state evolution as in Section 2. The complete posterior target is

$$p(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma \mid \boldsymbol{y}) \propto p(\boldsymbol{\theta}_0) \prod_{t=1}^{T} p(\boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1}) \prod_{t=1}^{T} p(y_t \mid \boldsymbol{\theta}_t, v_t, s_t, \sigma, \gamma)$$
$$\times \prod_{t=1}^{T} p(v_t \mid \sigma) \prod_{t=1}^{T} p(s_t)\, p(\sigma) \pi_\gamma(\gamma). \tag{31}$$

Conditional on $(\boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$,

$$\tilde{y}_t = y_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t, \qquad R_t = \sigma B_\gamma v_t, \tag{32}$$

which again gives a linear Gaussian state-space model for $\boldsymbol{\theta}_{0:T}$.

3.2 MCMC Updates

The current package default samples σ from its exact conditional given γ and then updates γ by bounded univariate slice sampling. Joint random-walk alternatives are also implemented, but the following target is the default exact-kernel path.

Latent v_t. Let $r_t = y_t - \boldsymbol{f}_t^\top \boldsymbol{\theta}_t - C_\gamma \sigma |\gamma| s_t$. Then

$$v_t \mid \cdot \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{r_t^2}{\sigma B_\gamma},\ \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \tag{33}$$

Latent s_t. Let $y_t^\circ = y_t - \boldsymbol{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $d_\gamma = C_\gamma \sigma |\gamma|$. Then

$$s_t \mid \cdot \sim \mathcal{N}^+(m_{s,t}, V_{s,t}), \qquad V_{s,t} = \left(1 + \frac{d_\gamma^2}{\sigma B_\gamma v_t}\right)^{-1}, \qquad m_{s,t} = V_{s,t} \frac{d_\gamma y_t^\circ}{\sigma B_\gamma v_t}. \tag{34}$$

State path. Given $(\boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma)$, sample $\boldsymbol{\theta}_{0:T}$ by FFBS using (32).

Scale. For fixed γ, let $y_t^\circ = y_t - \boldsymbol{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t$ and $u_t = C_\gamma |\gamma| s_t$. Then

$$\sigma \mid \cdot \sim \mathrm{GIG}(\lambda_\sigma, \chi_\sigma, \psi_\sigma), \tag{35}$$

where

$$\lambda_\sigma = -\left(a_\sigma + \frac{3T}{2}\right), \tag{36}$$

$$\chi_\sigma = 2b_\sigma + 2\sum_{t=1}^{T} v_t + \sum_{t=1}^{T} \frac{(y_t^\circ)^2}{B_\gamma v_t}, \tag{37}$$

$$\psi_\sigma = \sum_{t=1}^{T} \frac{u_t^2}{B_\gamma v_t}. \tag{38}$$

Skewness. The conditional kernel for γ is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma) \prod_{t=1}^{T} (\sigma B_\gamma v_t)^{-1/2} \exp\left[-\sum_{t=1}^{T} \frac{\{y_t - \boldsymbol{f}_t^\top \boldsymbol{\theta}_t - A_\gamma v_t - C_\gamma \sigma |\gamma| s_t\}^2}{2\sigma B_\gamma v_t}\right], \tag{39}$$

for $\gamma \in (L, U)$. The default sampler applies the bounded slice update in Section 6 directly to this kernel.

3.3 LDVB Approximation

The mean-field family is

$$q(\boldsymbol{\theta}_{0:T}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma) = q(\boldsymbol{\theta}_{0:T}) \prod_{t=1}^{T} q(v_t) \prod_{t=1}^{T} q(s_t)\, q(\sigma, \gamma). \tag{40}$$

The conjugate factors are updated conditionally on moments from $q(\sigma, \gamma)$. Define

$$\kappa_1 = \mathbb{E}\{1/(\sigma B_\gamma)\}, \qquad \kappa_2 = \mathbb{E}\{A_\gamma/(\sigma B_\gamma)\}, \qquad \kappa_3 = \mathbb{E}\{A_\gamma^2/(\sigma B_\gamma)\}, \tag{41}$$

$$\kappa_4 = \mathbb{E}\{C_\gamma |\gamma|/B_\gamma\}, \qquad \kappa_5 = \mathbb{E}\{\sigma C_\gamma^2 \gamma^2/B_\gamma\}, \qquad \kappa_6 = \mathbb{E}\{A_\gamma C_\gamma |\gamma|/B_\gamma\}. \tag{42}$$

Let $m_{1/v,t} = \mathbb{E}(v_t^{-1})$, $m_{s,t} = \mathbb{E}(s_t)$, and $m_{s^2,t} = \mathbb{E}(s_t^2)$. The state factor is Gaussian and is computed by Kalman smoothing with observation precision

$$\phi_t = \kappa_1 m_{1/v,t} \tag{43}$$

and pseudo-response

$$y_t^{\mathrm{VB}} = \frac{y_t \phi_t - \kappa_2 - \kappa_4 m_{s,t} m_{1/v,t}}{\phi_t}. \tag{44}$$

If $\eta_t = \boldsymbol{f}_t^\top \boldsymbol{\theta}_t$, write $m_{\eta,t} = \mathbb{E}(\eta_t)$, $S_{\eta,t} = \mathbb{V}(\eta_t)$, $m_{\Delta,t} = y_t - m_{\eta,t}$, and $S_{\Delta,t} = m_{\Delta,t}^2 + S_{\eta,t}$. Then

$$q(v_t) = \mathrm{GIG}\left(\frac{1}{2},\ \kappa_1 S_{\Delta,t} - 2\kappa_4 m_{\Delta,t} m_{s,t} + \kappa_5 m_{s^2,t},\ \kappa_3 + 2\mathbb{E}(\sigma^{-1})\right), \tag{45}$$

$$q(s_t) = \mathcal{N}^+(m_{s,t}^{\mathrm{VB}}, V_{s,t}^{\mathrm{VB}}), \tag{46}$$

$$V_{s,t}^{\mathrm{VB}} = \left(1 + \kappa_5 m_{1/v,t}\right)^{-1}, \tag{47}$$

$$m_{s,t}^{\mathrm{VB}} = V_{s,t}^{\mathrm{VB}} \left(\kappa_4 m_{\Delta,t} m_{1/v,t} - \kappa_6\right). \tag{48}$$

The nonconjugate factor $q(\sigma, \gamma)$ is approximated by a Laplace-Delta step on transformed coordinates

$$u = \log \sigma, \qquad z_\gamma = \mathrm{logit}\left(\frac{\gamma - L}{U - L}\right). \tag{49}$$

The transformed objective is the expected complete log-joint plus the Jacobian $u + \log\{d\gamma/dz_\gamma\}$. The package stores the transformed mode, the approximate covariance matrix, Delta-method moments, and ELBO components.

ELBO. The LDVB ELBO has the block decomposition

$$\mathcal{L}(q) = \mathcal{L}_{\text{like}} + \mathcal{L}_\theta + \mathcal{L}_v + \mathcal{L}_s + \mathcal{L}_{\sigma\gamma} + H\{q(\boldsymbol{\theta}_{0:T})\} + \sum_{t=1}^{T} H\{q(v_t)\} + \sum_{t=1}^{T} H\{q(s_t)\} + H\{q(\sigma,\gamma)\}. \quad (50)$$

The state and latent-variable blocks are exact for the current $q(\sigma, \gamma)$ moments; the (σ, γ) entropy and smooth expectations are evaluated under the Laplace-Delta approximation.

4 Static AL and exAL Regression with Ridge Priors

4.1 Model Hierarchy and Full Joint

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ have row vectors $\boldsymbol{x}_i^\top$, and let $\boldsymbol{\beta} \in \mathbb{R}^p$. The ridge/Gaussian coefficient prior is

$$\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{b}_0, \mathbf{B}_0), \qquad \boldsymbol{b}_0 \in \mathbb{R}^p, \quad \mathbf{B}_0 \in \mathbb{R}^{p\times p}. \quad (51)$$

Under the exAL likelihood,

$$v_i \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma), \qquad s_i \sim \mathcal{N}^+(0, 1), \quad (52)$$

$$y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i \sim \mathcal{N}\left(\boldsymbol{x}_i^\top\boldsymbol{\beta} + C_\gamma\sigma|\gamma|s_i + A_\gamma v_i,\ \sigma B_\gamma v_i\right). \quad (53)$$

The static AL model is obtained by setting $\gamma = 0$, removing s_i, and using A_0, B_0. The exAL full joint is

$$p(\boldsymbol{\beta}, \sigma, \gamma, \boldsymbol{v}, \boldsymbol{s}, \boldsymbol{y}) = p(\boldsymbol{\beta})p(\sigma)\pi_\gamma(\gamma)\prod_{i=1}^{n} p(y_i \mid \boldsymbol{\beta}, \sigma, \gamma, v_i, s_i)p(v_i \mid \sigma)p(s_i). \quad (54)$$

4.2 MCMC Updates

For fixed $(\sigma, \gamma, \boldsymbol{v}, \boldsymbol{s})$, define

$$y_i^\star = y_i - C_\gamma\sigma|\gamma|s_i - A_\gamma v_i, \qquad w_i = (\sigma B_\gamma v_i)^{-1}. \quad (55)$$

Let $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$ and $\boldsymbol{y}^\star = (y_1^\star, \ldots, y_n^\star)^\top$. Then

$$\boldsymbol{\Sigma}_{\beta|\cdot} = \left(\mathbf{B}_0^{-1} + \mathbf{X}^\top\mathbf{W}\mathbf{X}\right)^{-1}, \quad (56)$$

$$\boldsymbol{m}_{\beta|\cdot} = \boldsymbol{\Sigma}_{\beta|\cdot}\left(\mathbf{B}_0^{-1}\boldsymbol{b}_0 + \mathbf{X}^\top\mathbf{W}\boldsymbol{y}^\star\right), \quad (57)$$

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\boldsymbol{m}_{\beta|\cdot}, \boldsymbol{\Sigma}_{\beta|\cdot}). \quad (58)$$

Let $D_\gamma = C_\gamma|\gamma|$. The remaining exAL full conditionals are

$$s_i \mid \cdot \sim \mathcal{N}^+(m_{s,i}, V_{s,i}), \quad (59)$$

$$V_{s,i} = \left(1 + \frac{\sigma D_\gamma^2}{B_\gamma v_i}\right)^{-1}, \quad (60)$$

$$m_{s,i} = V_{s,i}\frac{D_\gamma(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta} - A_\gamma v_i)}{B_\gamma v_i}, \quad (61)$$

and, with $r_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - \sigma D_\gamma s_i$,

$$v_i \mid \cdot \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{r_i^2}{\sigma B_\gamma},\ \frac{A_\gamma^2}{\sigma B_\gamma} + \frac{2}{\sigma}\right). \tag{62}$$

For fixed γ, define $t_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - A_\gamma v_i$. Then

$$\sigma \mid \cdot \sim \mathrm{GIG}\left(-a_\sigma - \frac{3n}{2},\ 2b_\sigma + 2\sum_{i=1}^n v_i + \sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i},\ \sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i}\right). \tag{63}$$

The nonstandard γ kernel is

$$\pi(\gamma \mid \cdot) \propto \pi_\gamma(\gamma)\,[B_\gamma]^{-n/2} \exp\left\{\sum_{i=1}^n \frac{D_\gamma s_i t_i}{B_\gamma v_i} - \frac{1}{2\sigma}\sum_{i=1}^n \frac{t_i^2}{B_\gamma v_i} - \frac{\sigma}{2}\sum_{i=1}^n \frac{D_\gamma^2 s_i^2}{B_\gamma v_i}\right\}, \tag{64}$$

on (L, U). The package default updates this one-dimensional target by the bounded slice sampler in Section 6.

For the AL special case, the Gibbs updates reduce to

$$\boldsymbol{\beta} \mid \cdot \sim \mathcal{N}(\boldsymbol{m}_{\beta|\cdot}, \boldsymbol{\Sigma}_{\beta|\cdot}), \tag{65}$$

$$\sigma \mid \cdot \sim \mathrm{IG}\left(a_\sigma + \frac{3n}{2},\ b_\sigma + \sum_{i=1}^n v_i + \sum_{i=1}^n \frac{(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta} - A_0 v_i)^2}{2B_0 v_i}\right), \tag{66}$$

$$v_i \mid \cdot \sim \mathrm{GIG}\left(\frac{1}{2},\ \frac{(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2}{\sigma B_0},\ \frac{A_0^2}{\sigma B_0} + \frac{2}{\sigma}\right). \tag{67}$$

4.3 LDVB Updates and ELBO

The static exAL variational family is

$$q(\boldsymbol{\beta}, \boldsymbol{v}, \boldsymbol{s}, \sigma, \gamma) = q(\boldsymbol{\beta}) \prod_{i=1}^n q(v_i) \prod_{i=1}^n q(s_i) q(\sigma, \gamma). \tag{68}$$

The coefficient factor is Gaussian. With the same moment definitions as in (42),

$$\omega_i = \kappa_1 \mathbb{E}(v_i^{-1}), \tag{69}$$

$$\eta_i = y_i \kappa_1 \mathbb{E}(v_i^{-1}) - \kappa_4 \mathbb{E}(v_i^{-1}) \mathbb{E}(s_i) - \kappa_2, \tag{70}$$

$$\boldsymbol{\Sigma}_\beta = \left(\mathbf{B}_0^{-1} + \mathbf{X}^\top \operatorname{diag}(\omega_1, \ldots, \omega_n) \mathbf{X}\right)^{-1}, \tag{71}$$

$$\boldsymbol{m}_\beta = \boldsymbol{\Sigma}_\beta \left(\mathbf{B}_0^{-1} \boldsymbol{b}_0 + \mathbf{X}^\top \boldsymbol{\eta}\right), \tag{72}$$

where $\boldsymbol{\eta} = (\eta_1, \ldots, \eta_n)^\top$. The $q(v_i)$ and $q(s_i)$ updates are also closed form conditional on moments of $q(\sigma, \gamma)$. Let

$$\bar{r}_i = y_i - \boldsymbol{x}_i^\top \boldsymbol{m}_\beta, \qquad \overline{r_i^2} = \bar{r}_i^2 + \boldsymbol{x}_i^\top \boldsymbol{\Sigma}_\beta \boldsymbol{x}_i. \tag{73}$$

Then

$$q(v_i) = \mathrm{GIG}\left(\frac{1}{2},\ \kappa_1 \overline{r_i^2} - 2\kappa_4 \bar{r}_i \mathbb{E}(s_i) + \kappa_5 \mathbb{E}(s_i^2),\ \kappa_3 + 2\mathbb{E}(\sigma^{-1})\right), \tag{74}$$

$$q(s_i) = \mathcal{N}^+(m_{s,i}^{\mathrm{VB}}, V_{s,i}^{\mathrm{VB}}), \tag{75}$$

$$V_{s,i}^{\mathrm{VB}} = \left(1 + \kappa_5 \mathbb{E}(v_i^{-1})\right)^{-1}, \tag{76}$$

$$m_{s,i}^{\mathrm{VB}} = V_{s,i}^{\mathrm{VB}} \left\{\kappa_4 \bar{r}_i \mathbb{E}(v_i^{-1}) - \kappa_6\right\}. \tag{77}$$

The nonconjugate factor $q(\sigma,\gamma)$ is proportional to

$$q(\sigma,\gamma) \propto \pi_\gamma(\gamma)[B_\gamma]^{-n/2}\sigma^{-(a_\sigma+1+3n/2)}\exp\left\{-\frac{1}{2}\left(\frac{\bar{c}(\gamma)}{\sigma}+\bar{d}(\gamma)\sigma\right)+\bar{e}(\gamma)\right\}, \tag{78}$$

where

$$\bar{c}(\gamma) = 2b_\sigma + 2\sum_{i=1}^{n}\mathbb{E}(v_i) + \sum_{i=1}^{n}\frac{\mathbb{E}\{(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta} - A_\gamma v_i)^2/v_i\}}{B_\gamma}, \tag{79}$$

$$\bar{d}(\gamma) = \sum_{i=1}^{n}\frac{C_\gamma^2\gamma^2}{B_\gamma}\mathbb{E}\{s_i^2/v_i\}, \tag{80}$$

$$\bar{e}(\gamma) = \sum_{i=1}^{n}\frac{C_\gamma|\gamma|}{B_\gamma}\mathbb{E}\{s_i(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta} - A_\gamma v_i)/v_i\}. \tag{81}$$

This factor is approximated by the same Laplace-Delta method as in the dynamic exDQLM. The AL path sets $\gamma = 0$ and uses $q(\boldsymbol{\beta})q(\sigma)\prod_i q(v_i)$ with the same closed-form updates as Section 2, replacing state smoothing moments by $\mathbb{E}\{(y_i - \boldsymbol{x}_i^\top\boldsymbol{\beta})^2\}$.

The static exAL ELBO is

$$\mathcal{L}(q) = \mathbb{E}_q\{\log p(\boldsymbol{\beta})+\log p(\sigma)+\log\pi_\gamma(\gamma)\}+\sum_{i=1}^{n}\mathbb{E}_q\{\log p(y_i \mid \cdot)+\log p(v_i \mid \sigma)+\log p(s_i)\}+H(q), \tag{82}$$

where $H(q)$ is the sum of the Gaussian, GIG, truncated-normal, and Laplace-Delta entropy terms.

5 Static Regression with the RHS-NS Prior

This section focuses on the `rhs_ns` coefficient prior, the Gibbs-friendly regularized horseshoe variant used as the sparse-prior target for this supplement. It enters the static Gaussian update for $\boldsymbol{\beta}$ through a prior precision matrix. Let $\mathcal{A} \subseteq \{1,\ldots,p\}$ denote the active shrinkage set; by default an intercept, if present, is not shrunk. For $j \notin \mathcal{A}$ the package uses a fixed intercept precision. The direct `rhs` path is not developed here because its log-scale block is nonconjugate and is not the intended sparse-prior derivation for this supplement.

5.1 RHS-NS Prior: Package-exact Hierarchy

The `rhs_ns` option is the Gibbs-friendly regularized horseshoe variant used in the current implementation. For $j \in \mathcal{A}$,

$$\beta_j \mid \tau^2,\lambda_j^2,\zeta^2 \propto \mathcal{N}(\beta_j \mid 0,\tau^2\lambda_j^2)\mathcal{N}(\beta_j \mid 0,\zeta^2), \tag{83}$$

$$\lambda_j^2 \mid \nu_j \sim \mathrm{IG}(1/2,1/\nu_j), \qquad \nu_j \sim \mathrm{IG}(1/2,1), \tag{84}$$

$$\tau^2 \mid \xi \sim \mathrm{IG}(1/2,1/\xi), \qquad \xi \sim \mathrm{IG}(1/2,1/\tau_0^2), \tag{85}$$

$$\zeta^2 \sim \mathrm{IG}(a_\zeta,b_\zeta), \tag{86}$$

unless ζ^2 is fixed by the user. The resulting prior precision contribution for active coefficients is

$$\pi_j = \frac{1}{\tau^2\lambda_j^2}+\frac{1}{\zeta^2}. \tag{87}$$

Consequently, the $\boldsymbol{\beta}$ update in MCMC or VB is obtained from the static Gaussian update after replacing $\mathbf{B}_0^{-1}$ by a diagonal precision matrix with active entries π_j or their variational expectations.

5.2 RHS-NS MCMC Updates

Conditioning on $\boldsymbol{\beta}$, the shrinkage block has closed-form inverse-gamma updates:

$$\lambda_j^2 \mid \cdot \sim \mathrm{IG}\left(1,\ \frac{1}{\nu_j} + \frac{\beta_j^2}{2\tau^2}\right), \tag{88}$$

$$\nu_j \mid \cdot \sim \mathrm{IG}\left(1,\ 1 + \frac{1}{\lambda_j^2}\right), \tag{89}$$

$$\tau^2 \mid \cdot \sim \mathrm{IG}\left(\frac{|\mathcal{A}|+1}{2},\ \frac{1}{\xi} + \frac{1}{2}\sum_{j\in\mathcal{A}} \frac{\beta_j^2}{\lambda_j^2}\right), \tag{90}$$

$$\xi \mid \cdot \sim \mathrm{IG}\left(1,\ \frac{1}{\tau_0^2} + \frac{1}{\tau^2}\right), \tag{91}$$

$$\zeta^2 \mid \cdot \sim \mathrm{IG}\left(a_\zeta + \frac{|\mathcal{A}|}{2},\ b_\zeta + \frac{1}{2}\sum_{j\in\mathcal{A}} \beta_j^2\right), \tag{92}$$

with the last line omitted when ζ^2 is fixed.

5.3 RHS-NS VB Updates and ELBO Contribution

The package uses a mean-field approximation over the RHS-NS scale block. For $j \in \mathcal{A}$,

$$q(\lambda_j^2) = \mathrm{IG}(a_{\lambda j}, b_{\lambda j}), \qquad q(\nu_j) = \mathrm{IG}(a_{\nu j}, b_{\nu j}), \tag{93}$$

$$q(\tau^2) = \mathrm{IG}(a_\tau, b_\tau), \qquad q(\xi) = \mathrm{IG}(a_\xi, b_\xi), \tag{94}$$

$$q(\zeta^2) = \mathrm{IG}(a_\zeta^\star, b_\zeta^\star), \tag{95}$$

with $a_{\lambda j} = a_{\nu j} = a_\xi = 1$ and $a_\tau = (|\mathcal{A}|+1)/2$. Let $\bar\beta_j^2 = \mathbb{E}_q(\beta_j^2)$. The coordinate updates are

$$b_{\lambda j} = \frac{1}{2}\bar\beta_j^2\, \mathbb{E}(\tau^{-2}) + \mathbb{E}(\nu_j^{-1}), \tag{96}$$

$$b_{\nu j} = 1 + \mathbb{E}\{(\lambda_j^2)^{-1}\}, \tag{97}$$

$$b_\tau = \frac{1}{2}\sum_{j\in\mathcal{A}} \bar\beta_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1}), \tag{98}$$

$$b_\xi = \frac{1}{\tau_0^2} + \mathbb{E}(\tau^{-2}), \tag{99}$$

$$a_\zeta^\star = a_\zeta + \frac{|\mathcal{A}|}{2}, \qquad b_\zeta^\star = b_\zeta + \frac{1}{2}\sum_{j\in\mathcal{A}} \bar\beta_j^2. \tag{100}$$

The third line is shorthand for $b_\tau = \frac{1}{2}\sum_{j\in\mathcal{A}} \bar\beta_j^2 \mathbb{E}\{(\lambda_j^2)^{-1}\} + \mathbb{E}(\xi^{-1})$, where the expectation of $(\lambda_j^2)^{-1}$ is component-specific inside the summation.

The following term-by-term ELBO contribution matches the package's `rhs_ns` scale block. For $x \sim \mathrm{IG}(a, b)$ under the rate parameterization used above, define

$$m_{\log x}(a,b) = \log b - \psi(a), \qquad m_{x^{-1}}(a,b) = a/b, \qquad h_{\mathrm{IG}}(a,b) = a + \log b + \log\Gamma(a) - (a+1)\psi(a), \tag{101}$$

where $\psi(\cdot)$ is the digamma function and h_{IG} is the entropy of an inverse-gamma variational factor. Let

$$\ell_{\lambda j} = m_{\log x}(a_{\lambda j}, b_{\lambda j}), \qquad r_{\lambda j} = m_{x^{-1}}(a_{\lambda j}, b_{\lambda j}), \tag{102}$$

$$\ell_{\nu j} = m_{\log x}(a_{\nu j}, b_{\nu j}), \qquad r_{\nu j} = m_{x^{-1}}(a_{\nu j}, b_{\nu j}), \tag{103}$$

$$\ell_{\tau} = m_{\log x}(a_{\tau}, b_{\tau}), \qquad r_{\tau} = m_{x^{-1}}(a_{\tau}, b_{\tau}), \tag{104}$$

$$\ell_{\xi} = m_{\log x}(a_{\xi}, b_{\xi}), \qquad r_{\xi} = m_{x^{-1}}(a_{\xi}, b_{\xi}). \tag{105}$$

If ζ^2 is random, set $\ell_\zeta = m_{\log x}(a_\zeta^\star, b_\zeta^\star)$ and $r_\zeta = m_{x^{-1}}(a_\zeta^\star, b_\zeta^\star)$; if it is fixed at ζ_0^2, set $\ell_\zeta = \log \zeta_0^2$ and $r_\zeta = 1/\zeta_0^2$.

The coefficient-prior contribution is

$$\mathcal{L}_{\beta,\mathrm{hs}} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\log(2\pi) - \frac{1}{2}\ell_\tau - \frac{1}{2}\ell_{\lambda j} - \frac{1}{2}\bar{\beta}_j^2 r_\tau r_{\lambda j} \right\}, \tag{106}$$

$$\mathcal{L}_{\beta,\mathrm{slab}} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\log(2\pi) - \frac{1}{2}\ell_\zeta - \frac{1}{2}\bar{\beta}_j^2 r_\zeta \right\}. \tag{107}$$

The local, global, and slab prior contributions are

$$\mathcal{L}_{\lambda|\nu} = \sum_{j\in\mathcal{A}} \left\{ -\frac{1}{2}\ell_{\nu j} - \log\Gamma(1/2) - \frac{3}{2}\ell_{\lambda j} - r_{\nu j} r_{\lambda j} \right\}, \tag{108}$$

$$\mathcal{L}_{\nu} = \sum_{j\in\mathcal{A}} \left\{ -\log\Gamma(1/2) - \frac{3}{2}\ell_{\nu j} - r_{\nu j} \right\}, \tag{109}$$

$$\mathcal{L}_{\tau|\xi} = -\frac{1}{2}\ell_\xi - \log\Gamma(1/2) - \frac{3}{2}\ell_\tau - r_\xi r_\tau, \tag{110}$$

$$\mathcal{L}_{\xi} = \frac{1}{2}\log(\tau_0^{-2}) - \log\Gamma(1/2) - \frac{3}{2}\ell_\xi - \tau_0^{-2} r_\xi, \tag{111}$$

$$\mathcal{L}_{\zeta} = \begin{cases} a_\zeta \log b_\zeta - \log\Gamma(a_\zeta) - (a_\zeta+1)\ell_\zeta - b_\zeta r_\zeta, & \zeta^2 \text{ random}, \\ 0, & \zeta^2 \text{ fixed}. \end{cases} \tag{112}$$

The entropy contribution is

$$\mathcal{L}_{\mathrm{ent}} = \sum_{j\in\mathcal{A}} h_{\mathrm{IG}}(a_{\lambda j}, b_{\lambda j}) + \sum_{j\in\mathcal{A}} h_{\mathrm{IG}}(a_{\nu j}, b_{\nu j}) + h_{\mathrm{IG}}(a_\tau, b_\tau) + h_{\mathrm{IG}}(a_\xi, b_\xi) + \mathbb{I}(\zeta^2 \text{ random}) h_{\mathrm{IG}}(a_\zeta^\star, b_\zeta^\star). \tag{113}$$

Finally, when the intercept is not included in $\mathcal{A}$, the package adds the fixed Gaussian intercept-prior contribution

$$\mathcal{L}_{\beta_0} = \frac{1}{2}\{\log(\pi_0) - \log(2\pi)\} - \frac{1}{2}\pi_0 \bar{\beta}_0^2, \tag{114}$$

where π_0 is the fixed intercept precision. If the intercept is shrunk, this term is omitted because the intercept is already represented in the active RHS-NS block.

Thus the package contribution is

$$\mathcal{L}_{\mathrm{rhs\_ns}} = \mathcal{L}_{\beta,\mathrm{hs}} + \mathcal{L}_{\beta,\mathrm{slab}} + \mathcal{L}_{\lambda|\nu} + \mathcal{L}_{\nu} + \mathcal{L}_{\tau|\xi} + \mathcal{L}_{\xi} + \mathcal{L}_{\zeta} + \mathcal{L}_{\mathrm{ent}} + \mathcal{L}_{\beta_0}. \tag{115}$$

All terms are available in closed form because each scale factor is inverse-gamma. This is the RHS-NS component of the full static-model ELBO; the likelihood, latent-variable, and (σ, γ) terms are added by the surrounding static AL or exAL variational routine.

6 Bounded Slice Sampling and Laplace-Delta Blocks

6.1 Bounded Univariate Slice Sampler

Let $h(x)$ be an unnormalized log-density on a finite interval $[a, b]$, and let $x_0 \in (a, b)$. The bounded slice update implemented in the package is:

(1) Draw $e \sim \text{Exp}(1)$ and set the log-slice height $z = h(x_0) - e$.

(2) Draw $u \sim \text{Unif}(0, w)$ and initialize $L = x_0 - u$, $R = x_0 + w - u$.

(3) Step out left and right by increments of w, respecting the bounds a and b, until the interval endpoints lie below the slice or the maximum number of expansions is reached.

(4) Truncate the interval to $[a, b]$.

(5) Repeatedly draw $x_1 \sim \text{Unif}(L, R)$. If $h(x_1) \geq z$, accept x_1. Otherwise shrink the interval to $[L, x_1]$ if $x_1 > x_0$ and to $[x_1, R]$ otherwise.

For dynamic and static exAL MCMC fits, this sampler is applied to the one-dimensional γ target on (L, U) by default. The same bounded-slice utility is also available to internal one-dimensional nonconjugate blocks, but those legacy shrinkage targets are outside the scope of this supplement.

6.2 Laplace-Delta Approximation

For a nonconjugate block $\boldsymbol{\eta} \in \mathbb{R}^d$ with log target $\ell(\boldsymbol{\eta})$, let $\hat{\boldsymbol{\eta}}$ be a local mode and $\mathbf{H} = \nabla^2 \ell(\hat{\boldsymbol{\eta}})$. If $-\mathbf{H}$ is positive definite, the Laplace approximation is

$$q(\boldsymbol{\eta}) \approx \mathcal{N}\left(\hat{\boldsymbol{\eta}},\ [-\mathbf{H}]^{-1}\right). \tag{116}$$

For a smooth function $r(\boldsymbol{\eta})$, the Delta approximation used by LDVB is

$$\mathbb{E}\{r(\boldsymbol{\eta})\} \approx r(\hat{\boldsymbol{\eta}}) + \frac{1}{2}\,\mathrm{tr}\left[\nabla^2 r(\hat{\boldsymbol{\eta}})[-\mathbf{H}]^{-1}\right]. \tag{117}$$

This is used for smooth functions of (σ, γ) in dynamic and static exAL models.

7 Package-to-Algorithm Map

Function	Main algorithmic role
`exdqlmMCMC()`	Dynamic DQLM/exDQLM posterior simulation: FFBS for states, Gibbs for conjugate latent/scale blocks, and bounded slice for γ when exAL is used.
`exdqlmLDVB()`	Dynamic DQLM/exDQLM mean-field VB: CAVI for Gaussian/GIG/truncated-normal blocks and Laplace-Delta for (σ, γ) in exAL.
`exalStaticMCMC()`	Static AL/exAL posterior simulation: Gaussian coefficient update, Gibbs latent/scale updates, RHS-NS inverse-gamma scale updates when selected, and bounded slice for γ in exAL.
`exalStaticLDVB()`	Static AL/exAL mean-field VB: Gaussian coefficient factor, CAVI latent and RHS-NS scale factors, and Laplace-Delta for (σ, γ) in exAL.

8 Second-pass Verification Checklist

Before this supplement is submitted, perform the following checks:

(1) Compare every displayed DQLM expression against `DQLM-and-BQR---Theory/main.tex`.

(2) Compare every dynamic exDQLM expression against `univ-exDQLM---Ensemble/main.tex` and the current `exdqlmMCMC()` / `exdqlmLDVB()` implementation.

(3) Compare every static exAL ridge expression against `exAL---Regression/main.tex`.

(4) Verify the term-by-term RHS-NS ELBO against `static_beta_prior.R`.